MARKETING MATERIAL

GLOBAL COMMENT

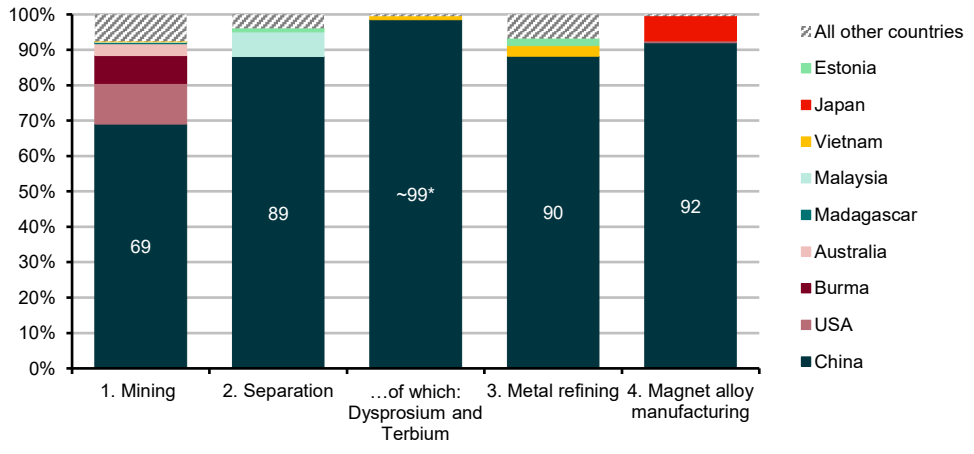
Rare earths likely to contain further escalation

China's dominance in rare earth elements (REEs) has been a concern since the early 2010s. In April 2025, REEs became a key negotiating lever in the US-China relationship. Despite longstanding efforts, US dependence on China persists. Even with ample capital and EBITDA guarantees, lead times stretch over years and are likely insufficient to ensure domestic US supply until 2028. Strategic stockpiles and recycling are inadequate. Thus, we believe China holds 'escalation dominance' for the next three to five years, likely steering the Trump administration toward further tactical concessions versus China despite its hawkish rhetoric.

We address the following questions in this report:

- What are critical raw materials, critical minerals, and REEs?
- How scarce are REEs really? Why are they critical? How is demand expected to increase?
- What is the REE value chain? How did China build its REE supply chain dominance?
- What are the key features of current Chinese export restrictions?
- What are the constraints to produce HREEs outside of China?
- MP Materials: Can capital alone accelerate homeshoring?
- How can the reliance on China be mitigated? Are there alternatives that require less HREEs?
- What are the implications for US-China tensions?

Global: Production volume of REEs in each step of value chain (%)



Source: US Department of Energy, USGS, Project Blue, Bloomberg, DNB Carnegie
Note: Lynas's dysprosium and terbium separation facility commissioned in Q2 2025 is not included, as volumes remain undisclosed

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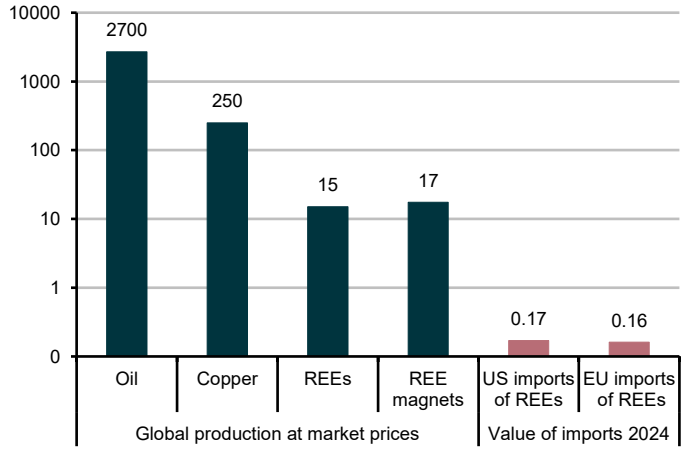
What are critical raw materials, critical minerals, and REEs?

Rare earth elements – also referred to as rare earths or REEs – are a subset of the broader category of critical raw materials (as designated by the EU) and critical minerals (as designated by the US). Both the EU and the US maintain official lists of critical raw materials/critical minerals, each covering roughly 50 materials. They are identified based on assessments of economic importance and supply risk (see appendix for full details). The lists are similar and include battery metals (such as lithium, cobalt, graphite, and nickel), more common base metals (such as copper, aluminium, zinc, and tin), and key minerals (such as gallium, germanium, and antimony). The latter three have been placed under Chinese export licencing requirements since 2023, followed by an export ban to the US in December 2024. None of these three are REEs, but are key minerals for semiconductor production. Although this may have supply chain implications, in this note, we only focus on the REEs.

REEs comprise 17 metallic elements that typically occur together in mineral deposits. Although not geologically rare, REEs are difficult to mine and refine. They seldom occur in concentrated deposits. More importantly, they are chemically similar, making separation technologically challenging. China controls most of the separation capacity today, and the related technical expertise.

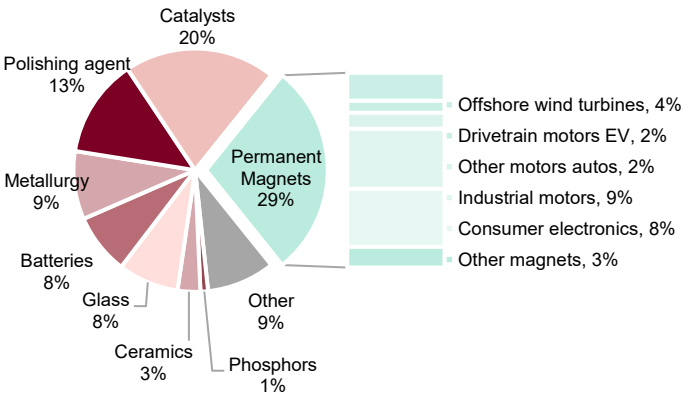
The market for REEs is very small compared with other key commodities (chart below left). They are used in small quantities but are vital, especially in the production of permanent magnets. Permanent magnets are used in offshore wind, electric vehicles (EVs), industrial motors, and consumer electronics (chart below right).

2024: Global market size (USDbn/year, in logarithmic scale)



Source: Bloomberg, USGC, EU Commission, DNB Carnegie

2020: Global civilian REE demand by sector (% of total)



Source: US Department of Energy, Oxford Institute of Energy Studies, DNB Carnegie
Note: The data only encompasses civilian commercial demand for rare earths. In the US, civilian commercial use of permanent magnets is 15–20x the volume of peacetime military use

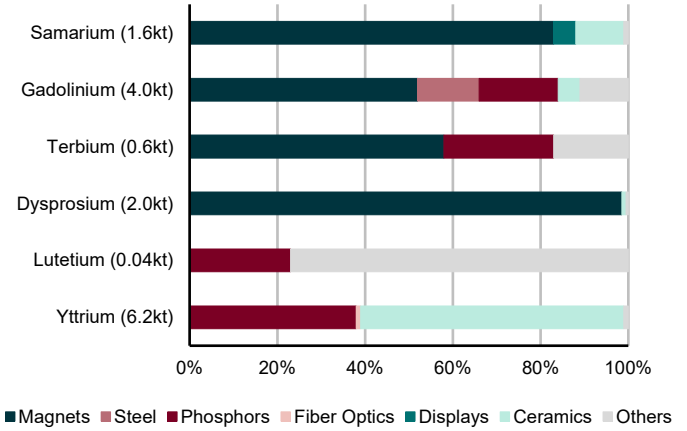
The 17 REEs are typically divided into two broad categories based on their atomic weight:

- **Seven Light Rare Earth Elements** (LREEs): Lanthanum, cerium, praseodymium, neodymium, promethium, samarium, and europium. Among the LREEs, neodymium and praseodymium are considered particularly critical due to their use in permanent magnets.
- **10 Heavy Rare Earth Elements** (HREEs): Gadolinium, terbium, dysprosium, holmium, erbium, thulium, ytterbium, lutetium, and yttrium. Scandium is sometimes included in this group. HREEs are relatively less abundant compared with the LREEs and are much more difficult to separate and refine. Among the most critical HREEs are dysprosium, terbium, and yttrium. Dysprosium enhances the thermal stability of neodymium-based magnets and is also used in nuclear reactors to help cool fuel rods to keep reactions under control. Terbium is used in display technologies, while yttrium is applied in screen manufacturing, specialty alloys, and chemical processes such as ethylene polymerisation.

HREEs are relatively less abundant than LREEs and are much more difficult to separate and refine

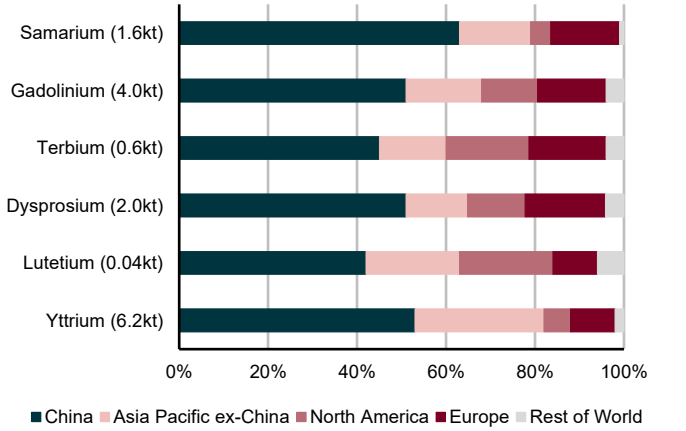
In April, China introduced export licencing requirements on seven such elements: yttrium, lutetium, dysprosium, terbium, gadolinium, samarium, and scandium. While they account for only 7% of global REE volumes, China controls up to 99% of the refining capacity for this subset – making the restrictions highly consequential for supply chains reliant on these inputs.

REEs under Chinese export control: Demand by sector (%)



Source: IEA Critical Minerals Outlook 2025, DNB Carnegie
Note: 2024 demand volumes 1000 tonnes (kt) included per REE

REEs under Chinese export control: Use by region (%)



Source: IEA Critical Minerals Outlook 2025, DNB Carnegie
Note: 2024 demand volumes 1000 tonnes (kt) included per REE

How scarce are REEs really?

REEs, despite their name, are not rare geologically…

Some may claim that the US securing access to REE exploration or mining rights in Ukraine would enhance supply security materially[1], but this does not align with the facts, in our opinion. Despite their name, REEs are not geologically rare. Cerium, for instance, is more abundant than copper and lead. Proven global reserves exceed current annual production by more than 400 times, and REE-bearing deposits are found on all populated continents. China, Russia, Brazil, and Vietnam together account for roughly 83% of proven reserves, with China alone estimated to hold between 35% and 50%. Australia now leads global REE exploration, accounting for 43% of spending on new mining projects, and Canada and Saudi Arabia are also expanding exploration efforts notably.

…but are technically complex and costly to process, making them vulnerable to supply-chain disruptions

However, geological availability does not equate to supply security. The real constraint lies in the ability to extract, separate individual REEs, and process them at scale. The REEs do not occur in pure form, but are chemically bound within various minerals deposits, often in low concentrations. Their nearly identical chemical properties make separation and purification complex and capital-intensive. Processing methods vary significantly across deposit types, affecting cost structures and environmental harm. In practice, separation processes must often be tailored to the specific mineralogy of each deposit, implying long lead times and the need for technical expertise.

Moreover, the real chokepoint in the REE supply chain lies not in access to the 17 elements as a group, but in the supply of specific ones. Although REE deposits are widespread, individual elements serve specialised purposes and are often difficult or impossible to substitute. It therefore makes little sense to assess REE availability in general terms. While some REEs are relatively abundant, others – particularly among the HREEs – are geologically scarcer, more geographically concentrated, and significantly more difficult and costly to refine. The most critical HREEs are extracted primarily from ion-adsorption clays found in southern China, and China controls both the upstream deposits and nearly all the downstream separation capacity. This is where China's position is especially dominant.

[1] Bloomberg 18.02.2025: Trump's Insistence Ukraine Has Rare-Earth Elements Is Wrong and Rare Earths in Ukraine? That's Utter Nonsense

Why are REEs critical?

In particular, NdFeB magnets are essential to motors used in EVs, wind turbines, and missile guidance systems

Permanent magnets are the most significant use of REEs: they convert motion into electricity and vice versa. NdFeB magnets, a key permanent magnet, are essential to high-performance motors, for instance those used in EVs, wind turbines, and precision missile guidance systems. While their base composition is neodymium (Nd), iron (Fe), and boron (B), REEs, such as dysprosium and terbium, are often added to improve thermal stability, helping the magnets retain their properties under high heat. Praseodymium is also used in NdFeB magnets.

By volume, the four key magnet REEs only represent 33% of the market, but in value, they comprise at least 90% of the market size of REEs. In end products, the NdFeB magnets represent a negligible share of costs, accounting for e.g. 0.5% of the EV cost. However, they are critical for performance, with few alternatives.

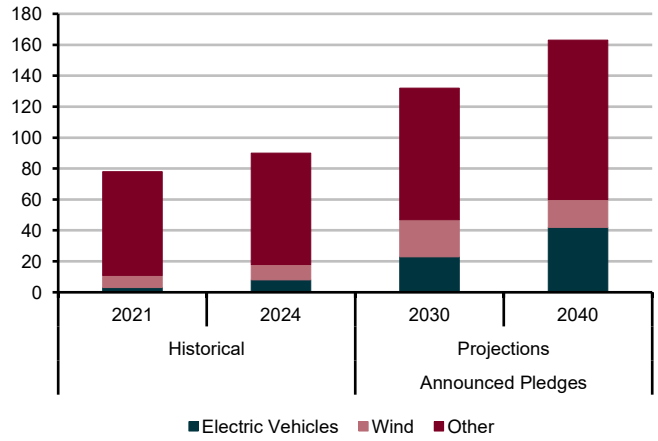
Furthermore, aircraft, drones, missiles, and munitions use rare earth actuators – small devices that control movement, like adjusting fins or flight surfaces. Although substitutes can be used in some applications, they are usually not as effective. According to the US 2021 Report on Stockpile Requirements, the volume of civilian commercial use of permanent magnets is 15–20x that of peacetime military use in the US, though the specifics are redacted. Each defence application requires a specially designed magnet with specific performance characteristics, suggesting to us it is not an easy task to quickly repurpose civilian commercial-use magnets.

How is the demand for REEs expected to increase?

Structural growth is largely thought to be driven by the demand for permanent magnets in EVs and offshore wind power

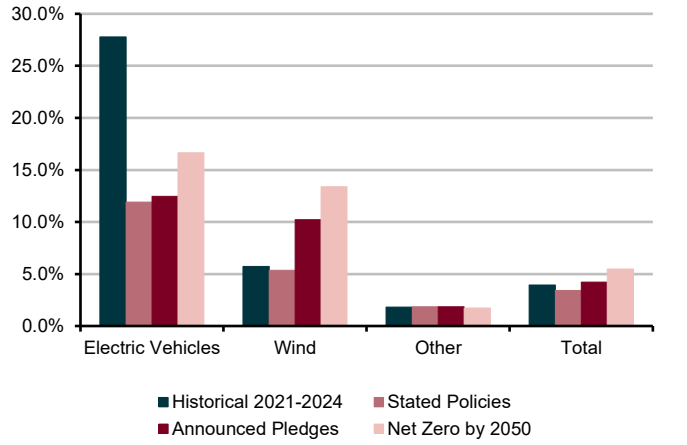
REEs are not facing the same supply/demand imbalance currently observed in critical raw materials such as graphite or copper, where future projected demand is outpacing current new supply investments. Nonetheless, demand for the four key magnet REEs is structurally increasing, driven primarily by their essential role in EV drivetrains[2] and offshore wind turbines[3] (see IEA estimates below). While these elements represent only a negligible fraction of total EV or wind turbine costs[4], their functional role is irreplaceable. Longer term, growth sectors such as industrial and humanoid robotics, may further amplify demand.

Magnet REEs: Demand by sector (kt per year)



Source: IEA Critical Minerals Outlook 2025 and DNB Carnegie
Note: The four magnet REEs are neodymium, praseodymium, dysprosium and terbium

Magnet REEs: Annual demand growth 2024–30 (CAGR %)



Source: IEA Critical Minerals Outlook 2025 and DNB Carnegie
Note: CAGR % refers to the implied compound annual growth rate for 2021–24, compared with the IEA's scenario-based projections for 2024–30. The 'Stated Policies' scenario reflects currently enacted government policies. The 'Announced Pledges' scenario incorporates additional measures aligned with countries' climate goals. The 'Net Zero by 2050' scenario represents a theoretical pathway consistent with achieving global net-zero CO_2 emissions by mid-century

[2] Most sources estimate that EV drivetrains use between one and two kilograms of NdFeB magnets. In addition to the motor, lesser grades of NdFeB magnets are also used in the motors for door locks, mirrors, seat positioning, power steering, suspension control, loudspeakers etc., making them key for the entire automotive industry.

[3] Although not all wind turbine systems require rare earth magnets, they are the preferred choice for offshore wind turbines due to reduced maintenance costs, generator efficiency, and generator weight.

[4] Global use of rare earth magnets in EVs may reach USD2.3bn–3.5bn by 2030, less than 0.5% of the value of the EV market, according to the European Raw Materials Alliance.

Going forward, China is the largest producer of REE magnets and a key global manufacturing hub for end-use products driving demand growth (EVs and offshore wind systems). As global magnet demand grows, we expect that international trade flows of REEs will increasingly occur in finished end-use goods (EVs and wind turbines) rather than the intermediate products (REE and magnets). This dynamic grants China a strategic advantage within end-use products that is not only based on price competitiveness, but also on controlling access to these functionally critical inputs.

What is the REE value chain?

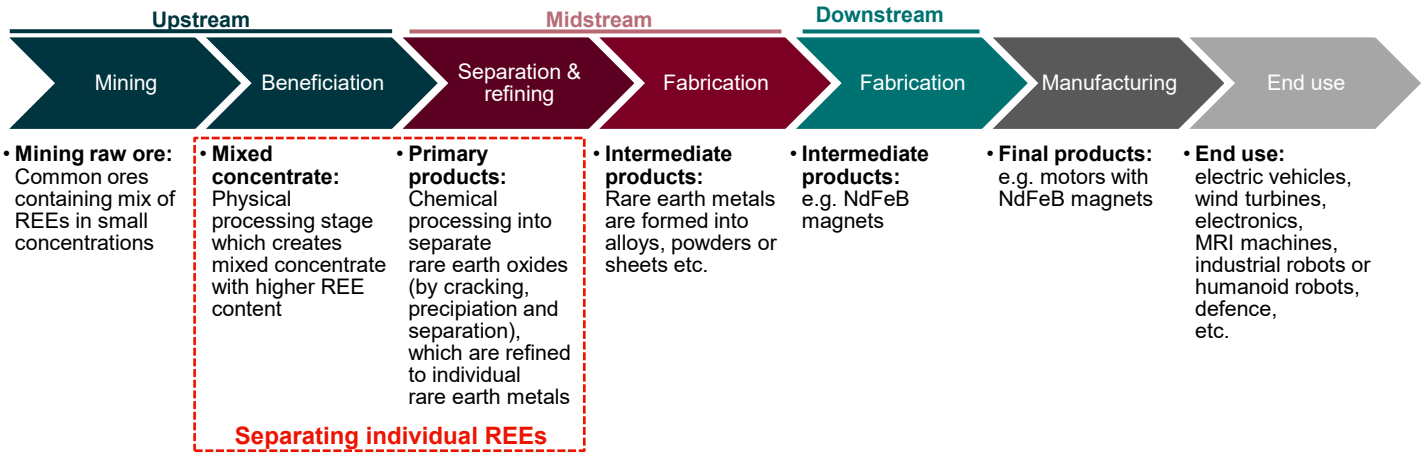
The rare earth supply chain is technically complex and resource-intensive. It depends on which REEs are targeted and the composition of the deposit and source rock. The type of deposit largely determines the type of REE extracted and the complexity of the subsequent refining process. There are two main types:

The type of deposit largely determines the type of REE extracted and the complexity of the subsequent refining process

- **Hard rock formations:** Most REEs are extracted from hard rock formations, such as carbonatites and granites, where they are chemically embedded in minerals such as bastnäsite[5], monazite[6], or xenotime[7]. These hard rock deposits dominate global supply and are the basis for production at major mines such as Bayan Obo in China and Mountain Pass in the US. Bastnäsite is most prevalent but yields primarily LREEs. Production from monazite is currently limited by the radioactive waste it yields, as it contains thorium. Xenotime is critical for heavy REEs but geologically scarce.
- **Ion-adsorption clays:** A second source of REEs is ion-adsorption clays, primarily found in southern China and, to a lesser extent, Myanmar. These deposits are particularly rich in HREEs, which do not form part of the mineral structures, but are instead loosely adsorbed onto the surface of clay minerals. They represent the main source of HREEs, with Chinese clay deposits alone accounting for more than 80% of global HREE production. Similar deposits are currently being explored in Australia, Brazil, Madagascar, Malawi, and elsewhere in Asia.

We will focus on the most critical parts of the value chain, which relate to the process of separating REEs. This involves beneficiation, separating, and refining (see figure below).

Simplified overview of the REE value chain



Source: IEA, USGS, Oxford Institute for Energy Studies, and DNB Carnegie analysis

[5] Bastnäsite, a carbonate mineral typically found in carbonatites, is especially rich in light REEs such as neodymium (Nd), praseodymium (Pr), cerium (Ce), and lanthanum (La).

[6] Monazite, a phosphate mineral found in igneous rocks and beach sands, also contains high levels of light REEs and some heavy REEs. However, its radioactivity – due to the presence of thorium and uranium – makes processing more costly and environmentally burdensome. As a result, several countries have restricted or abandoned its commercial use.

[7] Xenotime, another phosphate mineral, is richer in HREEs such as dysprosium (Dy), yttrium (Y), erbium (Er), and ytterbium (Yb). It is generally less radioactive than monazite and easier to leach, but economically viable deposits are limited.

Generally, extracting and processing REEs relies on similar steps, though it does vary depending on the resource type. For hard rock structures, these five stages are key:

- **Beneficiation:** The raw ore is mined then crushed and REE minerals are concentrated using physical methods such as magnetic separation, flotation, or gravity separation.[8] This increases REE concentration from 1–10% in raw ore to 50–60%. This step is typically done near the mine and requires basic mineral processing infrastructure.
- **Cracking (chemical digestion):** The concentrate undergoes chemical treatment to break down the mineral structure and dissolve the REEs into a liquid, known as a leach solution. This is the first step with significant environmental and regulatory implications, as acids or alkaline solutions are used.
- **Precipitation:** To separate rare earths from the liquid, a process called precipitation is used. Chemicals are added that cause the dissolved elements to form solid particles, usually as oxalates. These solids are then filtered out and collected for the next step. While relatively standardised, this step requires careful handling of chemical waste.
- **Separation:** Specific high-purity oxides (>99%) are isolated for each individual REE. This is the most important, complex, and capital-intensive stage. China currently dominates this step.

  Separation is the most important, complex, and capital-intensive stage. This is the key chokepoint

  The main method is solvent extraction, where a water-based rare earth solution is repeatedly mixed with an oil-based extractant that binds selectively to specific elements. Technicians must also fine-tune and adjust temperature and pressure, to allow the extractant to bind to the specific REE. Since REEs have nearly identical chemistry, the process must be repeated hundreds or thousands of times to reach target purity. China holds 70% of global patents in this area (WIPO, 2023). The specialised workforce to operate the separation facilities takes 5–7 years to train (National Science Foundation, 2025). Separation consumes large volumes of water, acids, and solvents, and generates significant chemical waste, an environmental barrier for many jurisdictions.

  China currently dominates this step, and is likely to continue to do so given its many decades of accumulated know-how
- **Refining:** Finally, the oxides are refined into rare earth metals through electrochemical processes. The process must be precise to avoid contamination and ensure the correct metal composition. This step enables the production of REE-based alloys, magnets, and other end-use materials and is often integrated directly with magnet production. China controls more than 90% of the global rare earth metal refining capacity.

Extraction from ion-adsorption clays bypasses many of the steps required for hard rock deposits. Because REEs in these clays are loosely bound to the surface of clay minerals, they can be extracted without crushing or high-temperature processing. Extraction is comparatively simple using mild salt solutions. However, recovery rates are low and environmental damage from poorly regulated operations has been significant.[9]

How did China build its REE supply chain dominance?

Until the mid-1980s, the US led global rare earth development in production and processing. The US pioneered separation techniques and was the top supplier of LREEs such as europium, used in colour displays for TVs and PC monitors. Activity centred on the Mountain Pass mine in California (chart below left). By the 1990s, environmental damage from radioactive waste and heavy metal leaks led to fines, and eventual closure. Mined production of REEs shifted to China, where environmental standards were much weaker and labour costs low.

The US dominated rare earth production until the mid-1980s, but China captured the market by the late 1990s

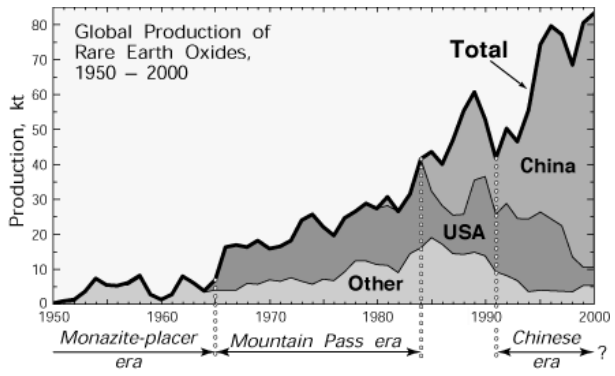
By 1999, more than 90% of the separated REEs used in the US came from China. Global mine production has risen sharply over the past decade – from 110kt in 2014 to 390kt in 2024. By

[8] In magnetic separation, magnets pull out minerals that respond to magnetic fields, leaving non-magnetic material behind. In flotation, crushed rock is mixed with water and chemicals, then aerated; rare earth minerals attach to bubbles and float, while waste sinks. The froth is skimmed and collected. Gravity separation relies on weight differences, as heavier rare earth minerals sink faster in water than lighter waste.

[9] Two main methods are used: topsoil removal, where clay-rich layers are stripped and leached with ammonium sulphate in collection ponds; and in-situ leaching, where chemicals are injected via boreholes and recovered through shallow pumps. The leachate is typically treated with oxalic acid to precipitate REEs. These low-cost methods require little infrastructure but pose serious environmental risks, including runoff, groundwater contamination, and long-term site degradation where environmental regulation is weak. Abandoned sites may leach pollutants for decades.

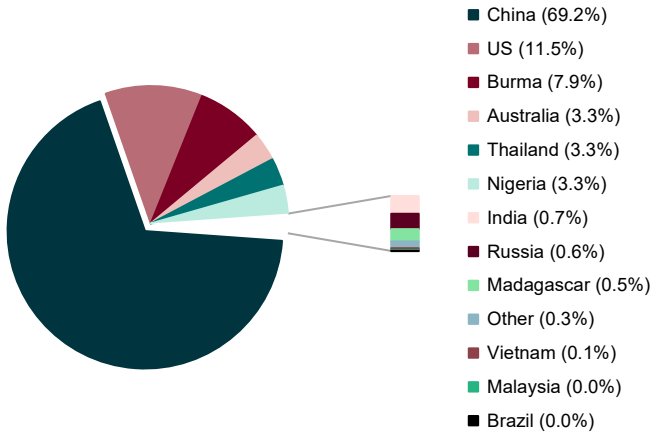
2024, China accounted for c70% of global mining production, totalling 270kt. In comparison, the US mined 45kt, representing only 11.5% of the global output (chart below right).

1950–2000: Mine production, by mine location not HQ (kt)



Source: USGS Fact Sheet (2002)
Note: HQ = headquartered

2024: Mine production, by mine location not HQ (%)



Source: USGS Mineral Commodity Summaries 2025 and DNB Carnegie
Note: HQ = headquartered. Chinese enterprises operate REE mines outside of China's borders

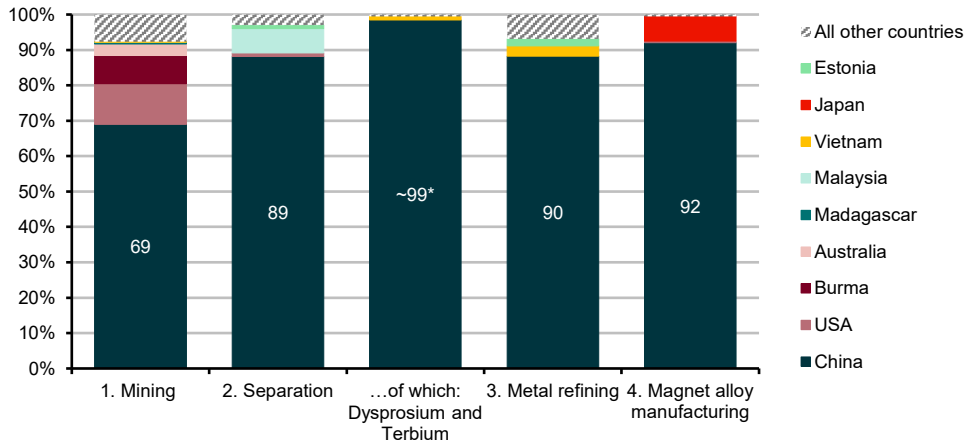
China's dominant position in rare earth mining and processing stems from a long-standing, coordinated state strategy to develop capabilities across every stage of the supply chain. The underlying logic being security of supply is only achieved when China has secured every independent stage of the supply chain. Key elements of China's policies include:

- Early exploration efforts, notably at Bayan Obo in Inner Mongolia in the 1950s, laid the foundation for industrial-scale mining, with China holding significant reserves.
- From the 1970s onwards, the Chinese government increased state-led investment in R&D, processing infrastructure, and export-oriented production.
- REEs were classified as "protected and strategic minerals" in 1990, limiting foreign access to upstream segments and enforcing joint ventures for downstream participation.
- During the 2000s, the government introduced export quotas and taxes, banned the export of rare earth concentrates, and implemented production limits to tighten control.
- As environmental degradation and illegal mining intensified, particularly in the south of China, the government initiated a consolidation of the sector. From 2010, the number of legal mines and processing facilities was drastically reduced. Consolidation has improved environmental oversight and reduced illegal operations, although challenges persist.
- By 2015, only six major state-backed enterprises were permitted to operate, a number further cut to four in 2021 with the creation of the China Rare Earth Group, which now controls 70% of domestic production. This vertical integration and consolidation have become key drivers of Chinese rare earths industrial competitiveness. Currently, China controls at least 89% of the production capacity in key steps of the value chain (chart below).

China has a long-standing strategy to ensure security of supply, meaning it needs to develop capabilities across every stage of the supply chain

By 2015, vertical integration and consolidation had become key drivers of Chinese rare earths industrial competitiveness

Global: Production volume of REE in each step of value chain (%)



Source: US Department of Energy, USGS, Project Blue, Bloomberg, DNB Carnegie
Note: Lynas's dysprosium and terbium separation facilities commissioned in Q2 2025 are not included as volumes remain undisclosed

Outside of China, there are only three separation facilities that produce rare earth oxides on an industrial scale

Notably, in May and June 2025, Lynas achieved an historic milestone, becoming the first company outside of China to produce commercial quantities of HREEs (dysprosium and terbium)

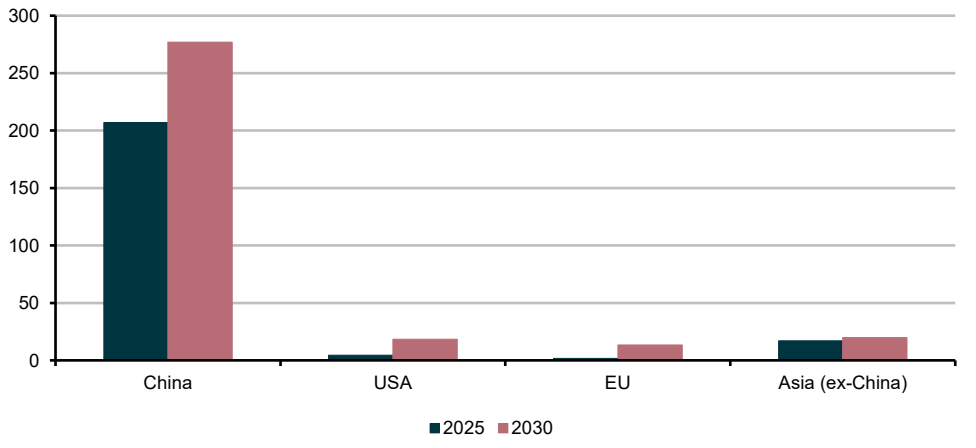
HREE separation: Lynas is a first-mover, but not a game-changer yet

Outside of China, only three industrial-scale REE separation plants exist: Mountain Pass (US), Silmet (Estonia), and Lynas's LAMP facility (Malaysia). In May–June 2025, Lynas became the first company outside of China to produce dysprosium and terbium oxides. This marked a technical milestone. However, actual output remains undisclosed and is likely modest. Despite nameplate capacity of 1.5kt/year, 2025 production is expected to reach just 30% of that amount. Constraints include limited feedstock and environmental challenges, particularly radioactive waste handling. Operational optimisation will take time. While symbolically important, Lynas's output covers only 3–6% of expected global HREE demand in 2025. Notably, the Trump administration's trade deal with Malaysia does not include an REE export clause.

Downstream magnet fabrication dominance shifted from US to Japan, before China

Downstream production of REE permanent magnets also shifted away from the US. By 1985, Japan had already exceeded the US in NdFeB magnet production, and by 1987 produced more than half of the world's magnets, before Chinese enterprises gained market share. Currently, downstream magnet production capacity is slowly scaling up to meet demand growth (chart below). However, based on announcements through April 2025, we believe China is likely to remain the dominant supplier for magnet production globally for the next five years, if not longer.

Global: Expected production capacity for NdFeB permanent magnets (kt per year)



Source: IEA analysis based on Project Blue and DNB Carnegie
Note: The list of projects is indicative of future magnet production capacity based on announcements through April 2025; it is not exhaustive

China is likely to remain the dominant supplier for magnet production globally for the next five years, if not longer, in our view

What are the key features of current Chinese export restrictions?

China's use of export controls has broadened steadily since 2023, covering a growing list of strategic materials and technologies, culminating in the 2025 rare earth licencing system.[10] On 4 April 2025, China activated a formal licencing system for the export of seven REEs and their related products, including metals, oxides, alloys, and magnets.

This builds on the 2020 Export Control Law, giving the Ministry of Commerce wide discretion to approve or reject shipments based on end-use and end-user declarations. In June, Beijing allowed exports to start flowing after the Trump administration agreed to a series of concessions (notably Nvidia H20 exports). However, China has maintained a lock on critical minerals for defence purposes. Exports are approved once the Chinese exporters can document to Chinese authorities that their products are not used by the US military industry.

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Unlike in 2010, when China briefly paused exports of REEs to Japan, which the WTO ruled against in 2014, this new export licencing system is based on national security. The system is nearly identical to President Biden's export controls on semiconductors, enforced by the BIS. For instance, China has similarly introduced extraterritorial restrictions on re-exports to the US military industry, and furthermore limited the exports not only of the REEs or permanent magnets, but also the required equipment and technology.

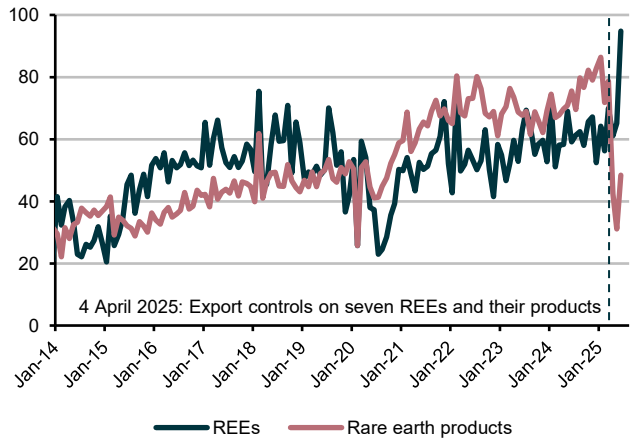
…however, the documentation process is difficult…

However, the documentation process is difficult. Chinese magnet suppliers may request the list of buyers or drawings of the final products to prove its exports would not go towards military applications.[11] Metal traders say that because China demands to know the end-user of rare earth magnets and metals, it is not approving licences for traders to stockpile. Of course, full control over re-exports of raw materials and intermediate products such as magnets is difficult, and we cannot rule out potential smuggling.

…and China is not approving licences for traders to stockpile REEs

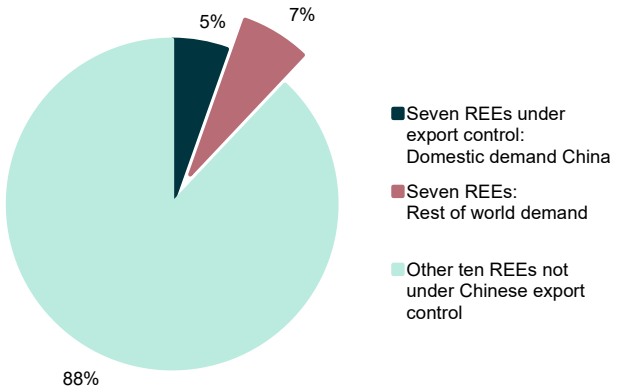
As shown in the chart below (left), exports of rare earth products fell sharply following the April announcement but partially recovered in June. At the same time, total REE export volumes surged to record levels. We believe this divergence likely reflects the targeted chokepoint on rare earth magnets now subject to licencing. In volume terms, the seven REEs under restriction account for only a small share of global demand (chart below right). In our view, the surge in exports of REEs likely reflects a wave of stockpiling, as buyers scramble to secure supplies of the 10 other REEs not under export controls ahead of potential further tightening.

China: Export volumes (kt per year, annualised*)



Source: LSEG Datastream, China Customs, DNB Carnegie
Note: *Seasonally adjusted by DNB Carnegie

Global: REEs demand (% of volumes)



Source: IEA and DNB Carnegie

[10] Key milestones include the introduction of export licencing for gallium and germanium in July 2023, materials critical to semiconductors, photovoltaics, and defence systems, marking the first application of China's 2020 Export Control Law. Between August and October 2023, China added rare earth magnet manufacturing technology, including sintered NdFeB processes, to its restricted technology list. In December 2023, licences were required for more than two dozen forms of graphite, particularly high-purity synthetic grades used in EV batteries and missile components. In 2024 and 2025, controls were extended to tungsten, covering powders, alloys, and carbides used in aerospace and defence. Most recently, in April 2025, export licencing was expanded to seven REEs, mainly heavy REEs, introducing a mechanism to block shipments based on end use and end user.

[11] Wall Street Journal 03.08.2025: China Is Choking Supply of Critical Minerals to Western Defense Companies

What are the constraints to produce the HREEs outside of China?

Although there are ten rare earths elements typically classified as HREEs, China's export restrictions introduced in April 2025 apply to only seven. In our view, there are three key constraints to producing these HREEs outside of China:

Geological scarcity outside of China

- First, deposits of HREEs are heavily concentrated in China. Most REE deposits, such as carbonatites, yield mainly LREEs, while HREEs occur in much smaller quantities. The primary commercial source of HREEs is ion-adsorption clay deposits, which are located almost exclusively in southern China, e.g. the clays of China's Zhaibei deposit. Although these clays account for just less than 3% of China's total REE reserves, they have supplied more than 80% of the world's production of HREEs. Certain elements such as dysprosium and terbium are currently extracted almost exclusively from these clays. While similar deposits have also been found in Australia, where the largest discovery was made by Northern Minerals, and the deep seabed muds off the coast of Japan, none have reached commercial scale.

Separation process is complex and faces severe environmental barriers

- Second, building up refining capacity for HREEs outside of China remains very challenging, and China controls 93–99% of capacity. HREEs are extremely difficult to separate, involving strong acids and toxic byproducts. Certain heavy rare earths can require more than 1,000 solvent extraction stages. Outside of China, this process is hindered by tight environmental regulations, lack of infrastructure, and limited technical expertise. Most of the US's legacy know-how was concentrated in LREE processing, not HREEs.

Technological know-how and human capital are important and 'locked-in'…

…with Chinese export bans on REE extraction and separation technologies

- Third, much of the critical know-how and technical expertise resides with Chinese engineers, and the government has moved to protect this advantage. In December 2023, Beijing imposed export bans on REE extraction and separation technologies, citing national security. This was a direct response to the Biden administration's 2022 CHIPS Act, which intended to choke off China's access to the future of AI by restricting exports of advanced semiconductors, required fabrication equipment, and semiconductor design software.

MP Materials: Can capital alone accelerate homeshoring?

Despite ample capital, the buildout of complex industrial infrastructure requires long lead times

The US Department of Defense (DoD) aims to establish a complete mine-to-magnet REE supply chain by 2027 to meet US national defence needs, and is set to require defence contractors to avoid Chinese imports. MP Materials, which operates the only active rare earths mine in the US (Mountain Pass), is central to this effort.

Mountain Pass was once the global leader in rare earths, supplying more than two-thirds of global demand in the 1980s. In 2020, it accounted for 15% of global REE output in concentrate form (38.5 kt/year), but lacked downstream refining capacity. MP Materials acquired Mountain Pass in 2017, with financial backing from Shenghe Resources, a company partly owned by the Chinese government. Shenghe Resources provided funding and secured an off-take agreement, becoming both a shareholder and the primary buyer of MP Materials' raw mining output.

Despite strong political support and substantial funding, in our view, full mine-to-magnet independence is unlikely until well after 2028. Texas is on track to become the US hub for rare earth processing and magnet manufacturing, but commissioning takes time.[12] MP Materials' own timeline illustrates the challenge: its new magnet manufacturing facilities backed by DoD financing guarantees are not set to begin commissioning until 2028.

Over the next 10 years, the DoD is set to commit to: (1) a price floor; (2) EBITDA guarantees; and (3) 100% off-take of all magnets produced at MP Materials' future '10X' facility. The facility is expected to begin commissioning in 2028, ramping up to full capacity over the course of a decade. However, once fully operational, 10X would have an annual capacity of 10kt of rare

[12] Since 2020, the DoD has funded MP Materials, Lynas USA, and Noveon to build LREE/HREE separation and magnet manufacturing. As of 2025, this midstream investment is concentrated in Texas: Lynas is constructing LREE and HREE facilities in Hondo and Seadrift; MP is building a metal, alloy, and magnet plant in Fort Worth; and Noveon is opening its first REE magnet factory in San Marcos. As of April 2025, there was still no operational HREE separation facility in the US.

earth magnets, or about 3% of China's current output (280 kt/year). In our view, this demonstrates that capital alone cannot accelerate homeshoring at scale.

More critically, while investments in mining and magnet production are progressing, the US continues to lack midstream capacity, particularly for HREE separation and refining. As of April 2025, there was still no operational HREE separation facility in the US, but the company is preparing to expand such capabilities at Mountain Pass. As for LREE separation, a refinery in California is established, and MP Materials reports that half of its production is refined there.

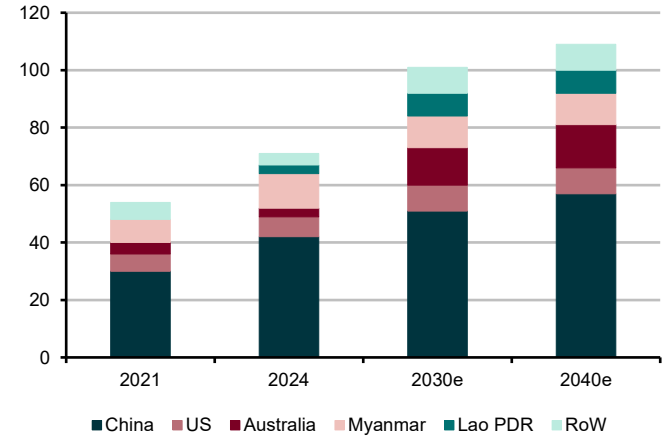
How can the reliance on China be mitigated?

The vulnerability of rare earth supply chains has been evident since 2010, when a dispute between China and Japan over the Senkaku/Diaoyu Islands temporarily cut off Japan's access to key REEs. However, since then, dependence on a single dominant supplier, China, has seemingly become more entrenched.

Coordinated Western efforts to localise REE supply chains have been underway since 2021, however success is not likely until 2030

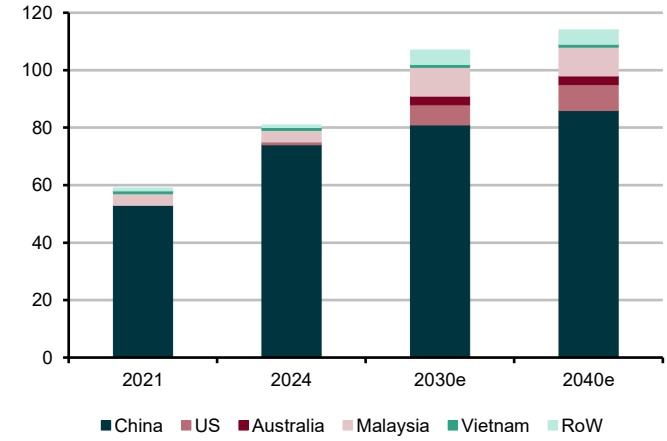
Western governments have taken increasingly coordinated steps since 2021 to reduce reliance on China.[13] Yet, we have seen limited actual shifts in supply chain geography and production capacity. China remains dominant in most stages of the value chain, retaining up 92–99% of refining capacity for the seven REEs currently under export control. Although mining capacity is set to diversify by 2030–40 (chart below left), refining capacity will likely stay highly concentrated in China (chart below right).

Four magnet REEs: Mining* (kt per year)



Source: IEA Critical Minerals Outlook 2025 and DNB Carnegie
Note: *Estimated production from operating and announced projects

Four magnet REEs: Refining* (kt per year)



Source: IEA Critical Minerals Outlook 2025 and DNB Carnegie
Note: *Estimated production from operating and announced projects

In the short term, stockbuild and improved recycling of REEs and rare earth magnets could in theory mitigate reliance on China.

Since China dominates magnet manufacturing, this means most scrap and swarf recovery occurs in China

- **Recycling:** Between 2021 and 2024, secondary supply (recycling and reuse) provided about 30% of total demand for the four key magnet REEs. However, most of the secondary recycled rare earth supply today comes from production scrap and leftover material from magnet manufacturing (scrap and swarf recovery), not from used or discarded products (end-of-life products). Since China dominates magnet manufacturing, this means most recycling (scrap and swarf recovery) occurs in China.

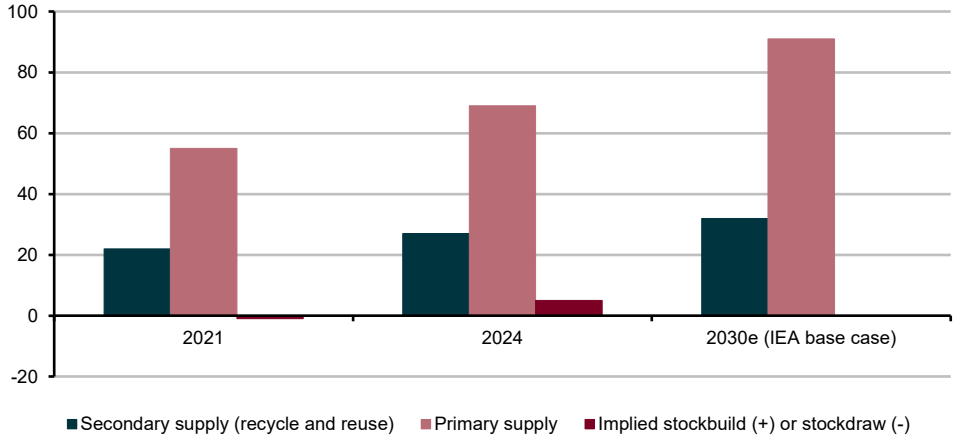
[13] In 2021, the US and EU launched formal supply chain reviews. In 2022, the US Inflation Reduction Act (IRA) tied tax credits to REE processing, while Canada, the EU, and Australia committed public funding to upstream rare earth mining and processing projects. In 2023, the EU Critical Raw Materials Act (CRMA) set binding targets for 2030: 10% of annual consumption to be extracted in the EU; 40% processed domestically; and 15% sourced from recycling, with no more than 65% of any strategic raw material coming from a single third country. Rare earths and permanent magnets are explicitly prioritised in the CRMA. In parallel, the Minerals Security Partnership, a US-led alliance with the EU, UK, Japan, Australia, Canada, and others which coordinates investment in non-Chinese mining and refining was established.

For many REEs, end-of-life recycling remains limited due to technical challenges and access to recycled feedstock. For example, technologies to recover yttrium from e-waste are still in early development, and while lanthanum and praseodymium can theoretically be extracted from discarded products, the process is inefficient, complex, and polluting. Several early-stage recycling initiatives are targeting NdFeB magnet recycling, though these are unlikely to materially shift supply dynamics during the remainder of the current US administration. With time, decommissioned offshore wind turbines may be eligible for increase end-of-life recycling.

For many REEs, end-of-life recycling remains limited due to technical challenges

In July 2025, MP Materials announced a strategic partnership with Apple to supply magnets made entirely from recycled materials, with shipments set to begin in 2027. Feedstock would include production scrap and end-of-life magnets. In our view, this reinforces the point that meaningful recycling of even low-end permanent magnets in consumer electronics is unlikely to scale until after 2027.

Four magnet REEs: Primary supply, recycling, and stockbuild (kt per year)



Source: IEA Critical Minerals Outlook 2025 and DNB Carnegie

Between 2021 and 2024, secondary supply – from recycling and reuse – met 30% of total demand for the four key magnet REEs…

…however, most of this comes from fabrication scrap and swarf, not from end-of-life products, meaning dependence on China for refined REEs remains

- **Commercial stockpiling:** In general, REEs with even atomic numbers tend to be more abundant in nature than those with odd numbers, making certain REEs less available than others. This natural imbalance is compounded by the fact that REEs typically occur in fixed ratios within the same mineral deposits. As a result, mining for a high-demand element such as neodymium inevitably yields larger volumes of more abundant, but less commercially valuable, elements such as lanthanum and cerium. Thus, absent a major increase, commercial stocks are unlikely to improve the security of supply for the scarcest REEs.
- **Strategic stockpiling:** Public data on strategic stockbuild is limited. For fiscal year 2023, the US Defense Logistics Agency (DLA) announced potential acquisitions of 100 tonnes of rare earth magnets, 0.6kt of neodymium, and 0.07kt of praseodymium, and has now expanded its stockpiling to include the HREEs dysprosium, samarium, and gadolinium. Although this may be enough to cover defence-related needs, the commercial civilian demand for REEs is significantly larger (15–20x). Going forward, we expect strategic stockpiling to increase as a short-term buffer against supply disruptions. Yet its effectiveness ultimately depends on the continued flow of Chinese exports.

Strategic stockpiling ultimately still depends on the continued flow of Chinese exports

Are there alternatives that require less HREEs?

In our view, the one long-term mitigation method would be to develop and adopt alternative technologies that reduce or eliminate the use of HREEs now subject to export controls. A similar shift occurred in the battery industry, where concerns over concentrated cobalt supply pushed manufacturers toward lithium-based chemistries. However, finding substitutes for rare earths in

Finding substitutes for rare earths in permanent magnets has proven difficult

permanent magnets has proven more difficult. As one industry expert put it: "The best time to start looking seriously into alternatives was 20 years ago. The second-best time is today"[14].

Europe is beginning a shift toward externally excited synchronous motors (EESMs), which use copper wire instead of permanent magnets in rotors. This approach reduces exposure to magnets in the automotive sector. S&P Global projects that European automakers' use of magnet-based motors could fall from 94% in 2025 to potentially 53% by 2035.

Another possible substitute is the MQ3 magnet, developed by General Motors in the 1980s and commercialised by Magnequench. These magnets use less HREE content while maintaining similar performance. However, MQ3 magnets are significantly more expensive to produce and would require major redesigns in downstream products, making rapid adoption unlikely, in our opinion.

What are the implications for US-China tensions?

In our view, China holds 'escalation dominance' in the rare earth value chain for at least the next five years. Its grip on midstream separation, refining, and fabrication gives China the ability to impose targeted export controls with minimal domestic economic impact.

We believe China holds 'escalation dominance' in the rare earth value chain for at least the next five years…

Total Chinese exports of all REEs was just USD490m in 2024, or 0.003% of Chinese GDP. Meanwhile, the US is left exposed across sectors, both commercial (such as automotive and clean tech) and defence (such as missiles, sensors, and motor systems). Not surprisingly, the US automotive sector expressed concern about the automotive supply chain, with Ford even temporarily shutting down its production line in Chicago in June 2025. European automakers were also caught in the crossfire.

Even as efforts have been made to boost capacity, such as Lynas's Malaysia facility or MP Materials' 10X project, the ramp up of actual capacity will take time. China's ability to restrict REE exports will be hard to match in the next few years.

In contrast, the US faces strategic trade-offs when considering escalation. For example, completely cutting off Nvidia's chip exports to China risks not only substantial lost revenue that funds Nvidia's R&D, but also long-term market share losses to rising rivals such as Huawei and Semiconductor Manufacturing International Corporation (SMIC). In practice, President Trump has already blinked: exports of Nvidia's chips appear to be continuing in exchange for access to Chinese REEs. China may well succeed in negotiating access to import high-bandwidth memory (HBM) chips[15]. Negotiations are likely to go over another 90 days, as the 90-day tariff pause set to expire 12 August.

Although Western efforts would diminish dependence on China for REEs, this will not happen overnight. This suggests to us that China holds 'escalation dominance' for the next three to five years. Given the transactional nature of the Trump administration thus far, this is likely steering the US toward tactical concessions, in our view, despite its hawkish rhetoric.

…despite tough rhetoric, this points toward President Trump likely adopting a bargaining approach when facing China

[14] The Economist 07.05.2025: How to build strong magnets without rare-earth metals
[15] Financial Times 10.08.2025: China wants US to relax export controls on chips as part of trade deal

Appendix

Overview of critical minerals and raw materials

Table: Critical minerals designated by the US; critical raw materials designated by the EU (REEs highlighted in grey)

List of US Critical Minerals (2022)

1. Aluminium
2. Antimony (**2024)
3. Arsenic
4. Barite
5. Beryllium
6. Bismuth (*2025)
7. Cerium
8. Cesium
9. Chromium
10. Cobalt
11. Dysprosium (*2025)
12. Erbium
13. Europium
14. Fluorspar
15. Gadolinium (*2025)
16. Gallium (**2024)
17. Germanium (**2024)
18. Graphite (*2023)
19. Hafnium
20. Holmium
21. Indium (*2025)
22. Iridium
23. Lanthanum
24. Lithium
25. Lutetium (*2025)
26. Magnesium
27. Manganese
28. Neodymium
29. Nickel
30. Niobium
31. Palladium
32. Platinum
33. Praseodymium
34. Rhodium
35. Rubidium
36. Ruthenium
37. Samarium (*2025)
38. Scandium (*2025)
39. Tantalum
40. Tellurium (*2025)
41. Terbium (*2025)
42. Thulium
43. Tin
44. Titanium
45. Tungsten (*2025)
46. Vanadium
47. Ytterbium
48. Yttrium (*2025)
49. Zinc
50. Zirconium

List of EU Critical Raw Materials (2023)

1. Antimony (*2024)
2. Arsenic
3. Baryte
4. Beryllium
5. Bismuth (*2025)
6. Boron
7. Cerium
8. Cobalt
9. Coking Coal
10. Copper
11. Dysprosium (*2025)
12. Erbium
13. Europium
14. Feldspar
15. Fluorspar
16. Gadolinium (*2025)
17. Gallium (*2024)
18. Germanium (*2024)
19. Graphite (*2023)
20. Helium
21. Holmium
22. Iridium
23. Lanthanum
24. Lithium
25. Lutetium (*2025)
26. Magnesium
27. Manganese
28. Neodymium
29. Nickel
30. Niobium
31. Osmium
32. Palladium
33. Phosphate Rock
34. Phosphorus
35. Platinum
36. Praseodymium
37. Promethium
38. Rhodium
39. Ruthenium
40. Samarium (*2025)
41. Scandium (*2025)
42. Silicon metal
43. Strontium
44. Tantalum
45. Terbium (*2025)
46. Thulium
47. Titanium metal
48. Tungsten (*2025)
49. Vanadium
50. Ytterbium
51. Yttrium (*2025)
52. Zinc

*= Under Chinese export licencing, by year introduced; ** = Under Chinese export bans, by year introduced (only applies to the US)

Source: European Commission, Congress.gov, IEA, and DNB Carnegie
Note: Promethium, a rare earth element, is only included in the EU list of critical raw materials. Molybdenum is not listed by either the US or EU, but has been subject to Chinese export licencing since February 2025. It has been highlighted by The Hague Centre for Strategic Studies 2023 as key for high-performance steels, but China only accounts for 38.7% of global supplies, compared with the US (23.3%) and Chile (14.9%).

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